

KYZYLORDA, Kazakhstan

WMO: 380620

Lat: 44.8247N

Lon: 65.5272E

Elev: 130

StdP: 99.77

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.1	-20.1	-26.5	0.3	-22.0	-23.8	0.4	-19.5	13.8	-13.1	12.2	-13.9	3.7	40	0.537

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.1	39.1	20.0	37.3	19.3	35.9	18.8	21.6	34.7	20.6	34.0	19.7	33.3	4.1	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.1	12.4	27.6	15.6	11.3	26.8	14.1	10.2	26.0	63.5	34.3	59.7	34.0	56.8	33.3	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	9.4	8.0	-25.6	41.9	4.3	1.7	-28.6	43.1	-31.1	44.1	-33.5	45.1	-36.6	46.4		
(5)			-25.5	23.5	5.0	2.5	-29.0	25.2	-31.9	26.7	-34.7	28.1	-38.3	29.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.1	-6.9	-4.5	4.5	13.5	20.7	25.8	27.3	25.6	18.7	10.6	2.1	-4.7		
	DBStd	13.28	6.51	7.17	6.31	5.42	4.01	3.56	3.16	3.66	4.39	5.07	5.84	6.11		
	HDD10.0	1905	525	408	188	28	1	0	0	0	1	55	243	456		
	HDD18.3	3532	783	641	429	159	23	1	0	2	48	244	487	714		
	CDD10.0	2318	0	0	18	133	331	474	536	484	263	73	6	0		
	CDD18.3	902	0	0	1	14	94	226	277	227	59	3	0	0		
	CDH23.3	12611	0	0	18	240	1253	3171	3959	3097	797	74	0	0		
	CDH26.7	6761	0	0	4	74	538	1768	2297	1734	331	16	0	0		
(14) Wind	WSAvg	3.4	3.4	3.3	3.8	3.9	3.9	3.3	3.2	3.5	3.2	2.9	3.0	3.3		
(15) Precipitation	PrecAvg	147	16	13	18	17	18	8	4	2	2	12	15	18		
	PrecMax	302	40	48	53	67	82	64	28	12	18	45	43	50		
	PrecMin	71	0	0	3	0	0	0	0	0	0	0	0	0		
	PrecStd	51	10	10	12	17	21	12	6	3	4	12	11	11		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.5	14.9	26.0	32.6	36.5	40.5	42.1	40.1	36.5	28.8	19.9	10.0		
		MCWB	4.9	8.6	14.2	17.4	17.9	20.4	20.9	20.3	18.6	15.0	12.3	5.4		
	2%	DB	4.7	9.4	20.8	28.8	33.8	38.2	39.8	38.1	33.0	25.2	15.8	6.9		
		MCWB	2.6	5.8	11.8	15.4	17.3	19.4	20.2	19.6	17.0	13.3	10.1	4.3		
	5%	DB	2.7	6.4	17.2	26.0	31.9	36.3	37.5	36.4	30.4	22.8	12.9	4.5		
		MCWB	1.3	3.5	9.9	14.2	16.4	19.0	19.5	19.0	16.0	12.4	8.0	2.6		
	10%	DB	1.0	4.1	14.2	23.5	29.8	34.7	35.7	34.7	27.9	19.9	10.1	2.6		
		MCWB	-0.2	1.7	8.5	13.1	15.9	18.5	19.0	18.4	15.1	11.0	6.5	1.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.3	9.8	14.5	18.5	21.0	23.2	23.3	22.6	19.2	15.6	13.4	6.7		
		MCDB	7.7	14.4	24.0	28.4	30.9	35.5	35.7	35.6	34.3	26.9	18.0	8.1		
	2%	WB	3.0	6.1	12.3	16.3	18.6	21.3	21.6	20.7	17.6	14.1	10.6	4.5		
		MCDB	4.4	8.8	19.2	25.7	29.8	33.6	35.1	35.4	31.2	23.7	14.9	6.5		
	5%	WB	1.4	3.8	10.5	15.0	17.5	20.1	20.6	19.8	16.5	12.7	8.4	2.8		
		MCDB	2.4	6.3	16.6	24.8	29.3	33.0	34.1	34.0	29.2	21.5	12.0	4.3		
	10%	WB	-0.1	1.9	8.7	13.6	16.5	19.2	19.7	18.9	15.4	11.3	6.5	1.2		
		MCDB	0.8	3.8	13.9	22.4	28.2	32.5	33.4	33.0	27.2	19.2	9.9	2.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	9.6	11.5	13.1	14.1	14.2	14.1	14.8	14.8	13.7	9.8	7.9		
		MCDBR	8.3	11.5	16.4	16.6	16.8	16.6	16.9	17.0	17.9	17.3	13.8	9.6		
		MCWBR	6.5	8.1	9.3	8.1	6.9	6.4	6.4	6.8	7.9	9.5	8.7	7.3		
		MCDBR	7.5	10.8	15.1	15.0	15.0	14.7	15.2	15.6	16.7	16.0	12.3	8.4		
	5% WB	MCWBR	6.1	7.9	9.1	7.9	7.1	6.9	6.3	6.8	7.9	9.1	8.2	6.7		
(40) Clear-Sky Solar Irradiance	taub		0.351	0.379	0.435	0.478	0.482	0.462	0.444	0.426	0.393	0.398	0.381	0.356		
	taud		2.066	2.033	1.981	1.912	1.949	2.056	2.130	2.154	2.188	2.094	2.101	2.085		
	Ebn at Noon		714	772	781	783	795	813	823	824	819	741	673	660		
	Edh at Noon		103	129	156	182	181	163	150	141	125	119	100	92		
(44) All-Sky Solar Radiation	RadAvg		1.41	2.58	3.94	5.44	6.91	7.48	7.35	6.63	5.21	3.33	1.88	1.24		
	RadStd		0.28	0.31	0.30	0.41	0.39	0.35	0.32	0.22	0.17	0.20	0.18	0.19		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	+1.18	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	+0.85	N/A	-1.08	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page